

Reuben Low Yu Xiang

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EDUCATION

Singapore Institute of Technology <i>Bachelor of Engineering in Robotics Systems Engineering (Honours with Distinction)</i>	Sep 2022 – May 2026
Singapore Polytechnic <i>Diploma in Mechatronics & Robotics, and Diploma Plus in Business</i>	Apr 2017 – Mar 2020

COURSEWORK

Robot Operating System (ROS1 & ROS2), Machine Learning, Reinforcement Learning, Robot Dynamics & Control, Computer Vision, Data Structures & Algorithms, Embedded Systems, System Engineering, Project Management

SKILLS

Languages: Python, C, C++
Frameworks & Tools: PyTorch, OpenCV, NVIDIA GR00T (VLA deployment), ROS1, ROS2, MoveIt2, NVIDIA Isaac Sim/Lab, Gazebo, RViz, Docker, Git/GitHub, Linux, MQTT, SolidWorks, Siemens NX, Visual Components
Hardware: Unitree H1, Unitree Go2, Kuavo V4/V4 Pro, NVIDIA Jetson AGX Orin, STM32, Arduino, Raspberry Pi, Intel RealSense, Omron TM5-700
Certifications: ROS-Industrial ROS2 Training; ROS-Industrial Consortium Asia Pacific (Sep 2025); Lufthansa-CAS Technical Training (TPE2018-1557)

WORK EXPERIENCE

Ceredroid AI Robot Learning and Motion Engineer	Jan 2025 – Dec 2025
- Developed autonomous manipulation and perception systems on the Kuavo humanoid and Unitree Go2 quadruped, spanning VLM-based task planning and evaluation, visual servoing, and vision-guided hardware payloads (Elaborated under Projects). - Collected teleoperated demonstrations on the Kuavo humanoid for NVIDIA GR00T VLA training, and deployed the trained policies on the robot. - Developed my capstone project in collaboration with the company: a Plan-Act-Correct-Verify (PACV) pipeline for long-horizon humanoid manipulation, using VLMs (Gemini Robotics ER 1.5, Cosmos-Reason1) as planners over VLAs (NVIDIA GR00T N1/N1.5). - Maintained and upgraded humanoid hardware (Kuavo, Unitree H1): replaced faulty motors, ran firmware updates and recalibration procedures, and authored maintenance SOPs adopted by the team. - Delivered interactive robotics demonstrations at SIT Open House, RoboSG, and community outreach events.	
SIT-FireFense Robotics Intern	Jun 2024 – Aug 2024
- Built an object detection pipeline for autonomous pick-and-place on an Omron TM5-700 manipulator, generating 3D pick targets from an Intel RealSense D435i depth camera. - Experimented with and integrated two detection backends into the pipeline, GPT-4o and a YOLO model. - Developed a Python driver for the Robotiq 2F-85 gripper via MODBUS protocol for the TM5-700.	
Singapore Institute of Technology Student Technical Lab Assistant	Jun 2023 – Dec 2024
- Provided technical support for robotics lab operations: soldering, wire crimping, lathe maintenance, and preparing materials and resources for robotics modules. - Performed equipment repair and maintenance on lab tools and robotic chassis used in coursework.	
Advanced Remanufacturing and Technology Centre (ARTC) Engineering Intern	Mar 2019 – Aug 2019
- Optimised mirror-finish polishing on stainless steel using the ABB IRB 140: experimented with spindle speeds, tool paths, and abrasive grits; analysed and documented surface roughness and visual quality. - Integrated ABB's Simplified Robot Programming (SRP) with the ABB IRB 6660; designed 3D-printed spray nozzle adapters and demonstrated motion tracking at the 2019 ROS-I APAC conference. - Developed components for a new cable management system on the ABB IRB 6660 robot arm, reducing cable strain and preventing entanglement.	

OTHERS

Singapore Armed Forces Supply Assistant	Jul 2020 – Jul 2022
Managed unit logistical operations. Awarded Best Soldier of the Month, Sergeant Major Coin, and SBW CO Letter of Commendation.	

Capstone: Hierarchical Long-Horizon Planning for Humanoid Robotics (Ceredroid AI)

Gemini Robotics ER 1.5, Cosmos-Reason1-7B, NVIDIA GR00T, Python

- Developed a long-horizon planning pipeline that decomposes natural language goals into subtasks, dispatches them to a **VLA**-driven Kuavo V4 Pro humanoid, visually verifies each step, and recovers from failures.
- Iterated from a stateless pipeline to a multi-turn conversation architecture for persistent dialogue context, and constrained VLM-generated plans to the robot's trained skill primitives to prevent hallucinated dispatch.
- Designed a comparative evaluation across 10,800 scored planning trials measuring task decomposition quality, sensitivity to scene complexity, and appropriate refusal of infeasible requests; Gemini ER 1.5 reached **97.2%** success rate vs **40.8%** for Cosmos-Reason1.
- Evaluated long-horizon monitoring on 50 **BEHAVIOR-1K** household tasks in simulation: decomposition (6.68/7), progress monitoring (2.01/3), with object hallucination the most common failure mode.

Transfer Learning for CIFAR-10 | *PyTorch, timm, DINOv3, CLIP, ResNet, Python*

- Compared three pre-trained frozen backbones on CIFAR-10, training only a 7,690-parameter head so the score reflects pre-trained feature quality: **DINOv3 ViT-B/16** reached **97.78%** against **94.87%** for **CLIP ViT-B/16**.
- Used the same data, split, seed and classifier head for both models, leaving the backbone as the only difference: **self-supervised** DINOv3 scored higher than **weakly-supervised** CLIP on identical **ViT-B/16** architecture.
- Analysed errors by per-class F1, confusion matrix and **t-SNE** feature projections: most residual error was cat and dog confusion, **126** cases for CLIP against **73** for DINOv3.

Storeman-Rosman: Autonomous Dual-Arm Shelf Picking (Ceredroid AI) | *Kuavo, ROS, YOLO, Python, REST*

- Implemented an autonomous dual-arm pick-and-place system on the Kuavo **humanoid**, with each arm independently fulfilling concurrent customer orders from a shared shelf.
- Defined function-call interfaces between the vision module and motion pipeline, dispatching trajectories from object detections through visual-servoing.
- Collected and annotated a canned-drink dataset across body and wrist camera views to fine-tune a **YOLO** detector, and enhanced the system with collision-avoidance trajectories and hard-limit safety callbacks for deployment.

Tic-Tac-Toe Humanoid Demo at RoboSG (Ceredroid AI) | *Kuavo, ROS, Computer Vision, SolidWorks, Python*

- Built an end-to-end demo in which the **Kuavo** played tic-tac-toe against the public, using a three-stage **ROS** pipeline (camera alignment, board classification, piece placement) with state management, command queuing, and blocking logic to prevent race conditions.

Autonomous Bird-Deterrence System (Ceredroid AI) | *Unitree Go2, Intel RealSense, YOLO, Arduino, SolidWorks, Python*

- Designed and built a vision-guided laser deterrence payload for the **Unitree Go2** quadruped, including the system's electrical schematic, integrating a **YOLO** model with **RealSense** depth sensing.
- Implemented **PID**-based target tracking that converts detections and depth measurements into pan-tilt angles, with multiple laser movement patterns to scare away birds.

Go2 Public Outreach Quadruped (Ceredroid AI) | *Android, MQTT, Jetson AGX Orin, SolidWorks, REST*

- Developed an Android teleoperation app for **Unitree Go2** integrating **MQTT** communication and the Unitree SportClient SDK; designed mounting hardware for the Jetson AGX Orin, speakers, and portable power to complete the deployable platform.

EASY Pastry Printer | *STM32, C, SolidWorks, Pixy Camera, SCRUM, Project Management*

- Collaborated with **Nicher**, an artisan bakery, to develop a food printer capable of printing 2D images and logos onto a tray of pastries in any orientation autonomously.

Bolloré Logistics | *ROS1, ROS2, UR10, AgileX-LIMO, Visual Components, STM32, C, Python, SolidWorks*

- Collaborated with **3PL** company Bolloré Logistics to produce a business case for a modular automated solution that allows for easy integration with emerging technologies for Value Added Service assembly lines.
- Simulated the solution in **Visual Components** and ROS 2 (Gazebo, RViz), then built and integrated a physical **proof-of-concept** of the assembly line (robotic platform and robotic arm).
- Business case projected a **22%** increase in throughput, from **6,807** to **8,352** kit boxes per 8-hour shift, with a **25%** reduction in operational costs and headcount and a projected **ROI period of three years**.